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Projected Rainfall Erosivity Under Climate Change in the Southeastern United States

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ABSTRACT. *Climate change is expected to change erosion rates as the erosive power of rainfall will most likely change due to changes in rainfall characteristics (e.g., intensity, duration, frequency, and subsequently energy). Since erosivity is greatly affected by rainfall intensity, it is expected that rainfall erosivity will be greatly altered as well. Few studies have estimated the impact of future climate changes on rainfall erosivity across the US or around the world. Furthermore, previously published erosion indices (based on historic data) have discrepancies due to the differences in methodologies (primarily due to omission of small and low rainfall intensity) adopted by those studies. Therefore, the objective of this study was to estimate changes in erosion indices for the period 2030–2059 using the benchmark rainfall indices established for the southeastern US by a recent study. Hourly precipitation data from 2030–2059 were retrieved for five climate models from the North-American Coordinated Regional Downscaling Experiment (NA-CORDEX)) datasets under the Representative Concentration Pathway (RCP) 8.5. The data was then temporally downscaled and bias-corrected using the quantile-based mapping method. The temporally downscaled and bias-corrected rainfall intensity data were then used to calculate future erosivity using WEPPCLIFF v1.6 and compared to the recently established erosivity benchmark to estimate expected changes in rainfall erosivity in the future. Our results found that average annual precipitation and rainfall erosivity are projected to increase by 14% and 47% respectively. Further, the findings indicate that the Gulf-Atlantic coast and Appalachian Mountains are most likely to be affected in the region.*

Keywords. *Erosivity, Climate Change, Rainfall Intensity, RCP8.5, RUSLE2, Soil Erosion.*

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INTRODUCTION

Soil erosion has been described as one of the biggest challenges for sustainable soil management by the United Nations Food and Agriculture Organization (FAO et al., 2019; Grillakis et al., 2020). Under changing climate, the future projections of soil erosion and its effects, including increased land degradation, decreased agricultural production, and losses of soil carbon are worsening (Borrelli et al., 2020; Panagos et al., 2022; Rahaman et al., 2022). The climate change is anticipated to cause soil erosion to increase by up to 60 % over the next century (Almagro et al., 2017; Foley et al., 2011; Hoomehr et al., 2016; Lapola et al., 2014; Yang et al., 2003). One of the primary driving forces for this process will likely be increasing temperatures coupled with changing rainfall characteristics, especially intensifying rainfall (Prein et al., 2017; Zhao et al., 2021). Therefore, quantifying the potential of rainfall to cause soil erosion through detachment and transport under changing climate is necessary (Fischer et al., 2018). Typically, point rainfall measurements are used to yield this potential, which is known as rainfall erosivity (Fischer et al., 2018; R. McGehee & Srivastava, 2018; Takhellambam et al., 2021; Wischmeier et al., 1965; Wischmeier & Smith, 1958, 1978).

The rainfall erosivity in the southeastern region of United States found between 2,000 to more than 10,000 MJ.mm.ha⁻¹h⁻¹yr⁻¹ (R. McGehee, 2016; R. McGehee & Srivastava, 2018). Moreover, the region has the biggest potential for shifting rainfall patterns (R. McGehee & Srivastava, 2018; Takhellambam et al., 2022). This necessitates to evaluate the rainfall erosivity to identify the potential risk under soil erosion. However, previous studies used aggregated (hourly or daily) rainfall data for estimation of future projected rainfall erosivity leading to underestimation of rainfall erosivity due to the smoothing of rainfall intensities (Fischer et al., 2018). In addition, previous erosion maps generated by Agricultural Handbooks (AH) of AH282, AH537, and AH703 were underestimated up to 30% when compared with the benchmark erosivity (McGregor et al. 1995; McGehee, 2016; McGehee and Srivastava, 2018).

Therefore, our objectives of our study are 1) quantifying the future projected rainfall erosivity using 15-min rainfall data for Southeast United States for 2030-59 under RCP8.5 scenario, and 2) Comparing the future projected erosivity with the historical period of 1971-2010. The findings from this study may be important for further estimation of soil erosion and related studies, especially for the Southeast United States.

MATERIAL AND METHODS

DESCRIPTION OF STUDY AREA AND DATA

The study area comprises the 11 states of Southeast United States (Figure 1). The states include Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, and Virginia. The observed average annual rainfall in this region varies from 1000 to 1500 mm and decreases as moved further inland from the coastal areas (Ingram et al., 2013; Kumar et al., 2021). However, average annual rainfall is expected to increase by the end of 2059 with more extreme rainfall events (Takhellambam et al., 2022). In this region, Gulf-Atlantic coast plays a vital key role for a peculiar climate country (Ingram et al., 2013; Kunkel et al., 2013).

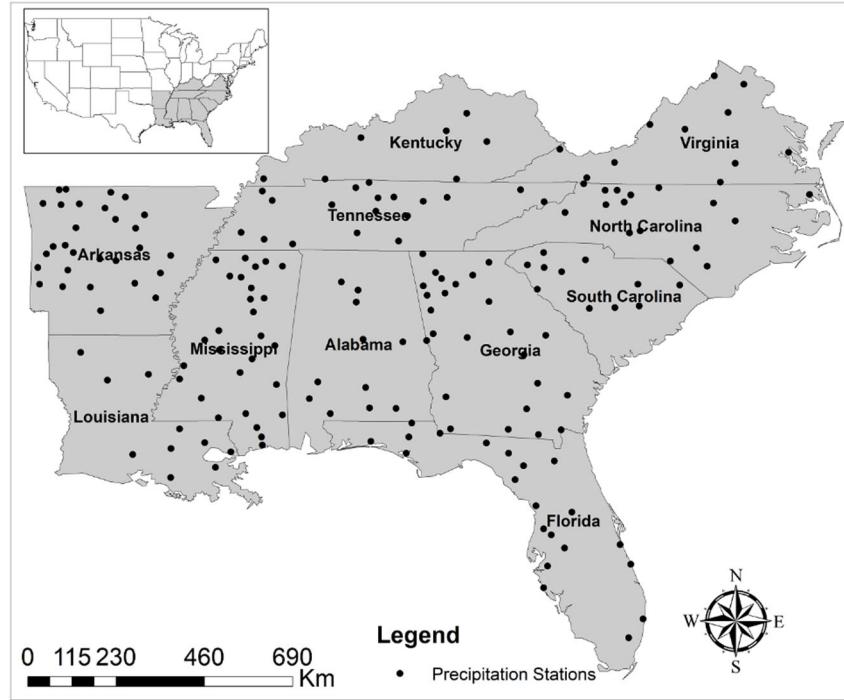


Figure 1. Map showing 187 rainfall stations that cover the entire 11 states of Southeast United States.

We obtained observed 15-min rainfall data from the National Oceanic and Atmospheric Administration for 528 stations for period of 1970 to 2013 (NOAA NCEI 2014). The datasets were further quality checked using a screening method known as 20.11. Please refer to McGehee and Srivastava (2018) for more details on this method. After screening, there are 187 remaining stations that were used for our study. In addition, 15-min rainfall data for both historical (1970-2010) and future (2030-59) period under five different climate models of North American-Coordinated Experiment Downscaling which were temporally downscaled from hourly using a modified stochastic approach (Takhellambam et al., 2022). General circulation models were simulated with forcing Coupled Model Intercomparison Project Phase 5 (CMIP5) representative concentration pathway (RCP) 8.5 having a spatial resolution of 50 km (Mearns et al., 2017; Scinocca et al., 2016). These five climate models are given as i) CANESM2_CANRCM4 (Canadian Earth System Model and Canadian Regional Climate Model version 4), ii) HadGEM2-ES.WRF (Hadley Centre Global Environment Model version 2 Earth system model and Weather Research and Forecasting), iii) GFDL-ESM2M.WRF (Earth System Model – Geophysical Fluid Dynamics Laboratory and Weather Research and Forecasting), iv) MPI-ESM-LR. RegCM4 (Max Planck Institute for Meteorology Earth System Model LR and Regional Climate Model version 4), and v) MPI-ESM-LR.WRF (Max Planck Institute for Meteorology Earth System Model LR and Weather Research and Forecasting) (Giorgi & Anyah, 2012; Scinocca et al., 2016; Skamarock et al., 2005).

EROSIVITY CALCULATIONS

The energy equation introduced by Revised Universal Soil Loss Equation version 2 (RUSLE2) is used through WEPPCLIFF v1.6. However, there are few update in the methodology as McGehee et al. (2021, 2022) found that the

same approach has discrepancies in the methodology due to the omission of smaller storms. Therefore, this study includes smaller storms and storm with more than 50-years of return period.

RESULTS AND DISCUSSION

PRECIPITATION

The average annual precipitation with 187 stations covering Southeast United States for observed, historical and future scenario is shown in Fig.2 below. The average precipitation for observed rainfall from 1970-2013 is found 1231 mm with minimum and maximum of 835 and 1689 respectively. Whereas, the historical period of 1970-99 shows the average annual precipitation of 1478-1742 under five RCM-GCMs.

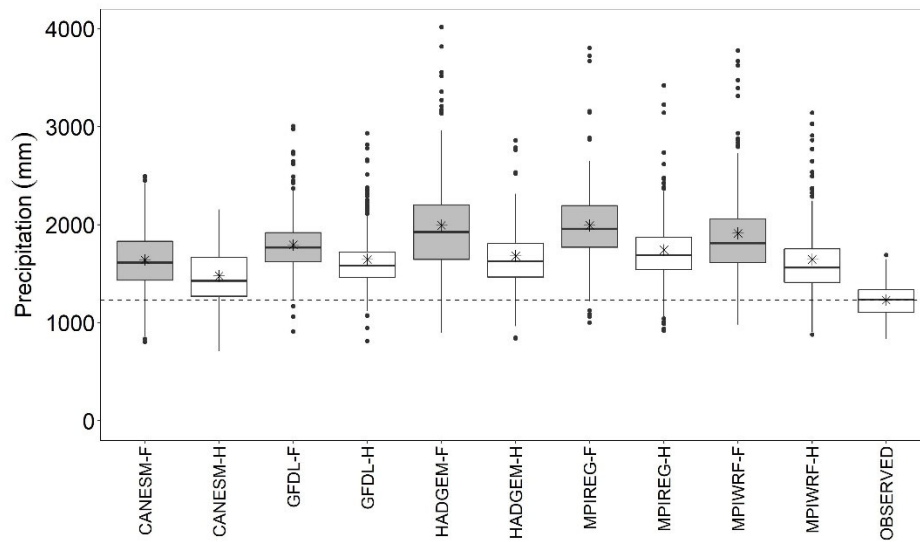


Figure 2. Boxplots showing the comparison of average annual rainfall between observed, historical and future scenario. Here, H and F represent historical and future scenarios respectively. In addition, ‘+’ indicates mean average value.

EROSIVITY INDEX

Table 1 shows the summary of the average annual rainfall erosivity for Southeast United States using 187 stations. The observed annual erosivity ranges from 1273 to 10587 MJ mm ha⁻¹h⁻¹yr⁻¹ during the period of 1971-2013 with a mean value of 4546 MJ mm ha⁻¹h⁻¹yr⁻¹. However, the historical model for five RCM-GCMs under RCP8.5 scenario for period of 1970-1999 shows between 1501 to 27286 MJ mm ha⁻¹h⁻¹yr⁻¹ with a mean range from 6527 to 8130 MJ mm ha⁻¹h⁻¹yr⁻¹. The main reasons for the overprediction of erosivity between the historical and observed values are mainly due to the bias correction and downscaling. This can be found from Takhellambam et al. (2022) that the average annual precipitation was overpredicted than the observed precipitation. Moreover, the temporal disaggregation of hourly to 15-min precipitation may not have similar rainfall characteristics with the observed rainfall. Considering this uncertainty, the future projected erosivity were compared with the historical period. The

average annual future (2030-59) erosivity were found in the range of 2112 to 41256 MJ mm ha⁻¹h⁻¹yr⁻¹ with mean value ranges from 9971 to 12985 MJ mm ha⁻¹h⁻¹yr⁻¹.

Further, Table 1 shows the relative increased in the average annual erosivity as compared to the historical period ranges from 23% to 61%. Likewise, the results using the ensemble model for both projected values of average annual precipitation and erosivity are found increase with 14% and 47% respectively. In the Figure 3, the higher amount of erosivity is found the both Gulf-Atlantic coast and Appalachian Mountain regions which is similar to the precipitation distribution. The main reasons for receiving greater amount of precipitation in these parts of the region are convective and orographic precipitation.

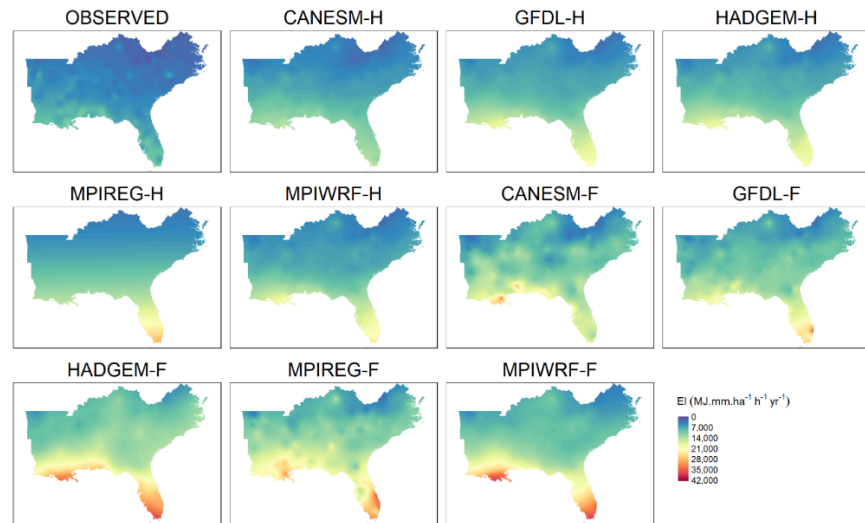


Figure 3. The spatial distribution of average annual Observed (1970-2013), historical (1970-1999) and future projected (2030-59) rainfall erosivity under RCP8.5 scenario over the Southeast United States.

Table 1: Statistical analysis of average annual erosivity using 187 stations for Southeast United states for observed (1970-2013), historical (1970-1999) and future (2030-59) period under RCP8.5 scenario for five RCM-GCMs.

Statistic	Observed	CANESM-H	GFDL-H	HADGEM-H	MPIREG-H	MPIWRF-H	CANESM-F	GFDL-F	HADGEM-F	MPIREG-F	MPIWRF-F
Min.	1273	1501	1686	1742	1807	1760	2186	2112	3174	2277	2419
Med.	4043	5806	6641	6975	6814	6272	9116	8546	10950	10995	8969
Max.	10587	15699	22439	19918	27286	23058	31408	35333	38125	35753	41256
Std. Dev.	1860	2748	3549	3566	4165	3953	4742	4283	6596	5788	6412
COV	0.41	0.42	0.45	0.44	0.51	0.51	0.48	0.44	0.51	0.47	0.57
Mean	4546	6527	7821	8075	8130	7773	9971	9655	12985	12366	11206
Obs. Abs. Diff.	0	1981	3275	3528	3583	3226	5425	5108	8438	7820	6660
Obs. Rel. Diff.	0%	44%	72%	78%	79%	71%	119%	112%	186%	172%	146%
Sim. Abs. Diff.							3444	1834	4910	4237	3433
Sim. Rel. Diff.							53%	23%	61%	52%	44%

CONCLUSION

We used WEPPCLIPP v1.6 and RUSLE2 energy approach for estimating future projected erosivity using RCP.5 scenario for the period of 2030-59 with five different climate models over the Southeast United States. The ensemble results showed the average annual precipitation and erosivity are projected to increase by 14% and 47% respectively. Gulf-Atlantic coast and Appalachian Mountain regions are likely to be affected the most as compared to the rest of the region. The results of this study will be bringing another dimension when we apply with a different approach for bias correction and downscaling. So, the future study is to bring an alternative approach. However, this is one of the first study that use high temporal resolution for future projected estimation of rainfall erosivity. Therefore, we affirm that there is likely to increase erosivity in the region with the changing climate in spite of the uncertainties associated with it.

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REFERENCES

- Almagro, A., Oliveira, P. T. S., Nearing, M. A., & Hagemann, S. (2017). Projected climate change impacts in rainfall erosivity over Brazil. *Scientific Reports*, 7(1), 8130. <https://doi.org/10.1038/s41598-017-08298-y>
- Borrelli, P., Robinson, D. A., Panagos, P., Lugato, E., Yang, J. E., Alewell, C., Wuepper, D., Montanarella, L., & Ballabio, C. (2020). Land use and climate change impacts on global soil erosion by water (2015-2070). *Proceedings of the National Academy of Sciences*, 117(36), 21994–22001. <https://doi.org/10.1073/pnas.2001403117>
- FAO, Clara, & Ronald. (2019). *Soil erosion: The greatest challenge for sustainable soil management* (p. 100).
- Fischer, F. K., Winterrath, T., & Auerswald, K. (2018). Temporal- and spatial-scale and positional effects on rain erosivity derived from point-scale and contiguous rain data. *Hydrology and Earth System Sciences*, 22(12), 6505–6518. <https://doi.org/10.5194/hess-22-6505-2018>
- Foley, J. A., Ramankutty, N., Brauman, K. A., Cassidy, E. S., Gerber, J. S., Johnston, M., Mueller, N. D., O'Connell, C., Ray, D. K., West, P. C., Balzer, C., Bennett, E. M., Carpenter, S. R., Hill, J., Monfreda, C., Polasky, S., Rockström, J., Sheehan, J., Siebert, S., ... Zaks, D. P. M. (2011). Solutions for a cultivated planet. *Nature*, 478(7369), 337–342. <https://doi.org/10.1038/nature10452>
- Giorgi, F., & Anyah, R. O. (2012). The road towards RegCM4. *Climate Research*, 52, 3–6. <https://doi.org/10.3354/cr01089>
- Grillakis, M. G., Polykretis, C., & Alexakis, D. D. (2020). Past and projected climate change impacts on rainfall erosivity: Advancing our knowledge for the eastern Mediterranean island of Crete. *CATENA*, 193, 104625. <https://doi.org/10.1016/j.catena.2020.104625>
- Hoomehr, S., Schwartz, J. S., & Yoder, D. C. (2016). Potential changes in rainfall erosivity under GCM climate change scenarios for the southern Appalachian region, USA. *CATENA*, 136, 141–151. <https://doi.org/10.1016/j.catena.2015.01.012>
- Ingram, K., Dow, K., Carter, L., & Anderson, J. (2013). *Climate of the southeast united states: Variability, change, impacts, and vulnerability*. Island Press. <https://link.springer.com/book/10.5822/978-1-61091-509-0>
- Kumar, H., Srivastava, P., Otiz, B. V., Morata, G., Lamba, J., & Bondesan, L. (2021). *FIELD-SCALE SPATIAL AND TEMPORAL SOIL WATER VARIABILITY IN IRRIGATED CROPLANDS*. 37. <https://doi.org/10.13031/trans.14335>
- Kunkel, K. E., Stevens, L. E., Stevens, S. E., Sun, L., Janssen, E., Wuebbles, D., Ii, C. E. K., Fuhrman, C. M., Keim, B. D., Kruk, M. C., Billet, A., Needham, H., Schafer, M., & Dobson, J. G. (2013). *Regional Climate Trends and Scenarios for the U.S. National Climate Assessment* (NESDIS 142-2; p. NESDIS 142-2). National Oceanic and Atmospheric Administration.
- Lapola, D. M., Martinelli, L. A., Peres, C. A., Ometto, J. P. H. B., Ferreira, M. E., Nobre, C. A., Aguiar, A. P. D., Bustamante, M. M. C., Cardoso, M. F., Costa, M. H., Joly, C. A., Leite, C. C., Moutinho, P., Sampaio, G., Strassburg, B. B. N., & Vieira, I. C. G. (2014). Pervasive transition of the Brazilian land-use system. *Nature Climate Change*, 4(1), 27–35. <https://doi.org/10.1038/nclimate2056>
- McGehee, R. (2016). *Development of Reliable Erosion Indices for Climate-Informed Soil Conservation in the Southeastern United States*. <https://etd.auburn.edu/handle/10415/5459>
- McGehee, R. P., Flanagan, D. C., Srivastava, P., Engel, B. A., Huang, C.-H., & Nearing, M. A. (2022). An updated isoerodent

- map of the conterminous United States. *International Soil and Water Conservation Research*, 10(1), 1–16. <https://doi.org/10.1016/j.iswcr.2021.06.004>
- McGehee, R. P., Flanagan, D. C., Srivastava, P., & Nearing, M. A. (2021). Rainfall erosivity: Essential historical, conceptual, and practical perspectives for continued application. In *Precipitation*. https://scholar.google.com/citations?view_op=view_citation&hl=en&user=JBKG7N8AAAAJ&sortby=pubdate&citation_for_view=JBKG7N8AAAAJ:Tyk-4Ss8FVUC
- McGehee, R., & Srivastava, P. (2018). Benchmarking reliable erosion indices from quarter-hour station data for climate studies in the southeastern United States. *Journal of Soil and Water Conservation*, 73(4), 363–376. <https://doi.org/10.2489/jswc.73.4.363>
- Mearns, L., McGinnis, S., Korytina, D., Arritt, R., Biner, S., Bukovsky, M., Chang, H.-I., Christensen, O., Herzmann, D., Jiao, Y., Kharin, S., Lazare, M., Nikulin, G., Qian, M., Scinocca, J., Winger, K., Castro, C., Frigon, A., & Gutowski, W. (2017). *The NA-CORDEX dataset* [NetCDF]. UCAR/NCAR. <https://doi.org/10.5065/D6SJ1JCH>
- NOAA, Administration, National Centers for Environmental, & Information). (2014). *US 15 Minute Precipitation Data, Version 1.0. 1970-2010*. Boulder, CO: NOAA National Centers for Environmental Information.
- Panagos, P., Borrelli, P., Matthews, F., Liakos, L., Bezak, N., Diodato, N., & Ballabio, C. (2022). Global rainfall erosivity projections for 2050 and 2070. *Journal of Hydrology*, 610, 127865. <https://doi.org/10.1016/j.jhydrol.2022.127865>
- Prein, A. F., Rasmussen, R. M., Ikeda, K., Liu, C., Clark, M. P., & Holland, G. J. (2017). The future intensification of hourly precipitation extremes. *Nature Climate Change*, 7(1), 48–52. <https://doi.org/10.1038/nclimate3168>
- Rahaman, H., Rehman, S., Singh, R., Ahmed, R., & Sajjad, H. (2022). *State of Art of Review on Climate Variability and Water Resources: Bridging Knowledge Gaps and the Way Forward*. 49(4), 12.
- Scinocca, J. F., Kharin, V. V., Jiao, Y., Qian, M. W., Lazare, M., Solheim, L., Flato, G. M., Biner, S., Desgagne, M., & Dugas, B. (2016). Coordinated Global and Regional Climate Modeling. *Journal of Climate*, 29(1), 17–35. <https://doi.org/10.1175/JCLI-D-15-0161.1>
- Skamarock, W., Klemp, J., Dudhia, J., Gill, D., Barker, D., & Wang, W. (2005). *A Description of the Advanced Research WRF Version 2* (p. 2500 KB) [Application/pdf]. UCAR/NCAR. <https://doi.org/10.5065/D6DZ069T>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R., & Kumar, H. (2021, December 13). *Potential changes in rainfall erosivity under climate change in southeastern United States*. AGU Fall Meeting 2021. <https://agu.confex.com/agu/fm21/meetingapp.cgi/Paper/989740>
- Takhellambam, B. S., Srivastava, P., Lamba, J., McGehee, R. P., Kumar, H., & Tian, D. (2022). *Temporal disaggregation of hourly precipitation under changing climate over the Southeast United States*. 14. <https://doi.org/10.1038/s41597-022-01304-7>
- Wischmeier, W. H., 1911-, Smith, D. D., & 1905-. (1965). *Predicting rainfall-erosion losses from cropland east of the Rocky Mountains*. <https://agris.fao.org/agris-search/search.do?recordID=US201300420250>
- Wischmeier, W. H., & Smith, D. D. (1958). Rainfall energy and its relationship to soil loss. *Transactions, American Geophysical Union*, 39(2), 285. <https://doi.org/10.1029/TR039i002p00285>
- Wischmeier, W. H., & Smith, D. D. (1978). *Predicting Rainfall Erosion Losses: A Guide to Conservation Planning*. Department of Agriculture, Science and Education Administration.
- Yang, D., Kanae, S., Oki, T., Koike, T., & Musiak, K. (2003). Global potential soil erosion with reference to land use and climate changes. *Hydrological Processes*, 17(14), 2913–2928. <https://doi.org/10.1002/hyp.1441>
- Zhao, W., Kinouchi, T., & Nguyen, H. Q. (2021). A framework for projecting future intensity-duration-frequency (IDF) curves based on CORDEX Southeast Asia multi-model simulations: An application for two cities in Southern Vietnam. *Journal of Hydrology*, 598, 126461. <https://doi.org/10.1016/j.jhydrol.2021.126461>